Lab Assignment 7: The MAC Sublayer

1. Decode by hand (5 marks)

Binary of the Frame Control bytes:

- **F1:** `08 01` (hex)
  - Byte 0: `08` to `0000 1000` (Protocol Version 00, Type 10 (Data), Subtype 0000 (Data))
  - Byte 1: `01` to `0000 0001` (ToDS 1, FromDS 0, all other flags 0)
- **F3:** `b4 00` (hex)
  - Byte 0: `b4` to `1011 0100` (Protocol Version 00, Type 01 (Control), Subtype 1011 (RTS))
  - Byte 1: `00` to `0000 0000` (All flags 0)

Field	F1	F3
Type / Subtype	Data / Data	Control / RTS
ToDS, FromDS	1, 0	0, 0
Other flags set	None	None
Duration (µs)	44	350
Address 1 (role)	00:1a:2b:3c:4d:5e (BSSID)	00:1a:2b:3c:4d:5e (RA)
Address 2 (role)	a4:83:e7:11:22:33 (SA)	a4:83:e7:11:22:33 (TA)
Address 3 (role)	00:50:56:aa:bb:cc (DA)	Not present
Sequence no. / fragment no.	467 / 0	Not present

2. Parser (7 marks)

The python parser is implemented in `parser.py`. Below is the printed summary output when running the script for frames F1 to F6:

```
=== F1 ===
Type/Subtype : Data / Data
Flags        : ToDS
Duration     : 44 µs
Addresses    :
  - BSSID: 00:1a:2b:3c:4d:5e
  - SA: a4:83:e7:11:22:33
  - DA: 00:50:56:aa:bb:cc
Seq / Frag   : 467 / 0
EtherType    : 0x0800

=== F2 ===
Type/Subtype : Data / Data
Flags        : FromDS, Retry
Duration     : 44 µs
Addresses    :
  - DA: a4:83:e7:11:22:33
  - BSSID: 00:1a:2b:3c:4d:5e
  - SA: 00:50:56:aa:bb:cc
Seq / Frag   : 468 / 0
EtherType    : 0x0800

=== F3 ===
Type/Subtype : Control / RTS
Flags        : None
Duration     : 350 µs
Addresses    :
  - RA: 00:1a:2b:3c:4d:5e
  - TA: a4:83:e7:11:22:33
```

```
=== F4 ===
Type/Subtype : Control / CTS
Flags        : None
Duration     : 324 µs
Addresses    :
  - RA: a4:83:e7:11:22:33

=== F5 ===
Type/Subtype : Control / ACK
Flags        : None
Duration     : 0 µs
Addresses    :
  - RA: a4:83:e7:11:22:33

=== F6 ===
Type/Subtype : Data / Data
Flags        : ToDS, FromDS, More Frag
Duration     : 44 µs
Addresses    :
  - RA: 02:11:22:33:44:55
  - TA: 02:66:77:88:99:aa
  - DA: 00:50:56:aa:bb:cc
  - SA: a4:83:e7:11:22:33
Seq / Frag   : 469 / 1
EtherType    : 0x0800
```

C3. Interpretation (3 marks)

1. F3, F4 and F5 are one exchange. Name the sender and receiver of each, and explain why the Duration drops from F3 to F4 to F5.

- **F3 (RTS):** Sender = `a4:83:e7:11:22:33`, Receiver = `00:1a:2b:3c:4d:5e`
- **F4 (CTS):** Sender = `00:1a:2b:3c:4d:5e`, Receiver = `a4:83:e7:11:22:33`
- **F5 (ACK):** Sender = `00:1a:2b:3c:4d:5e`, Receiver = `a4:83:e7:11:22:33`

The Duration field drops because it specifies the total time remaining to complete the entire frame exchange, updating at each step to protect subsequent frames:

- In **F3 (RTS)**, duration (350 µs) covers CTS + SIFS + DATA + SIFS + ACK.
- In **F4 (CTS)**, duration (324 µs) covers the remaining DATA + SIFS + ACK.
- In **F5 (ACK)**, duration (0 µs) is zero because the exchange is now complete.

2. A station that hears only F4 sets its NAV. For how many microseconds, and which slide mechanism does this implement?

A station hearing F4 sets its NAV (Network Allocation Vector) for **324 microseconds** (as specified in the Duration field of the CTS). This implements the **Virtual Carrier Sensing** mechanism (part of CSMA/CA). It solves the hidden terminal problem by ensuring that nodes that cannot hear the original sender (but can hear the receiver's CTS) know the medium will remain busy for the upcoming DATA and ACK frames.

3. F2 has the Retry flag set. How does the receiver use the Retry bit together with the sequence number to discard duplicates?

When the receiver gets a frame with the **Retry bit set (1)**, it indicates the frame is a retransmission. The receiver checks the Sequence Number (468) and Fragment Number (0) of this incoming frame against a cache of sequence numbers it has recently successfully received from this specific Source Address. If there is a match in the cache, the receiver knows it has already processed this exact packet. It will silently discard the duplicate frame to avoid passing it to the higher layers, but it will still send an ACK back to the sender (so the sender stops retransmitting).

4. Why does F6 need a fourth address? Sketch the path it takes, using the WDS figure from the slides.

F6 has both ToDS = 1 and FromDS = 1, which is characteristic of a **Wireless Distribution System (WDS)** or mesh frame forwarded between two Access Points over a wireless backbone. Since the wireless link itself is a bridge between two other distinct segments, four addresses are necessary to identify the immediate endpoints of the wireless hop and the ultimate source/destination:

- **RA (Receiver Address):** `02:11:22:33:44:55` (The receiving AP of this hop)
- **TA (Transmitter Address):** `02:66:77:88:99:aa` (The transmitting AP of this hop)
- **DA (Destination Address):** `00:50:56:aa:bb:cc` (The ultimate destination STA)
- **SA (Source Address):** `a4:83:e7:11:22:33` (The original source STA)

Path sketch:

```
[Original STA] --(wireless/wired)--> [Transmitting AP] ==(WDS wireless link)==> [Receiving AP] --(wireless/wired)--> [Final STA]
     (SA)                                 (TA)                                      (RA)                                (DA)
```